

KPI Definitions

This document defines the business KPIs used by the ESG Analytics Assistant. It is intended for Retrieval-Augmented Generation (RAG) to provide business context during SQL generation and insight generation.

Scope 1 Emissions

Business Definition: Direct greenhouse gas emissions from owned or controlled sources.

Recommended Aggregation: SUM

Unit: Metric Tons CO₂e

Business Interpretation: Higher values indicate greater direct operational emissions.

Typical Drill-down: Country, Industry, Facility, Fiscal Year

Recommended Visualization: Bar, Trend Line

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Scope 2 Emissions

Business Definition: Indirect emissions from purchased electricity.

Recommended Aggregation: SUM

Unit: Metric Tons CO₂e

Business Interpretation: Used to evaluate electricity-related carbon impact.

Typical Drill-down: Country, Industry, Facility

Recommended Visualization: Bar, Trend Line

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Scope 3 Emissions

Business Definition: Indirect value-chain emissions.

Recommended Aggregation: SUM

Unit: Metric Tons CO₂e

Business Interpretation: Often the largest contributor to total carbon footprint.

Typical Drill-down: Country, Supplier, Facility

Recommended Visualization: Bar, Trend Line

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Carbon Footprint

Business Definition: Combined Scope1 + Scope2 + Scope3 emissions.

Recommended Aggregation: SUM

Unit: Metric Tons CO₂e

Business Interpretation: Overall greenhouse gas footprint.

Typical Drill-down: Country, Industry, Facility

Recommended Visualization: Stacked Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Energy Consumption

Business Definition: Energy consumed across facilities.

Recommended Aggregation: SUM

Unit: MWh

Business Interpretation: Measures operational energy usage.

Typical Drill-down: Country, Industry, Facility

Recommended Visualization: Bar, Line

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Renewable Energy Percentage

Business Definition: Share of energy from renewable sources.

Recommended Aggregation: AVG

Unit: Percentage

Business Interpretation: Higher values indicate cleaner energy adoption.

Typical Drill-down: Country, Industry

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Water Withdrawal

Business Definition: Total water withdrawn.

Recommended Aggregation: SUM

Unit: Million Litres

Business Interpretation: Tracks freshwater dependency.

Typical Drill-down: Country, Facility

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Water Recycled Percentage

Business Definition: Share of withdrawn water recycled.

Recommended Aggregation: AVG

Unit: Percentage

Business Interpretation: Higher values indicate better sustainability.

Typical Drill-down: Country, Facility

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Waste Generated

Business Definition: Waste produced by operations.

Recommended Aggregation: SUM

Unit: Metric Tons

Business Interpretation: Measures waste output.

Typical Drill-down: Facility, Industry

Recommended Visualization: Horizontal Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Waste Recycled Percentage

Business Definition: Percentage of waste recycled.

Recommended Aggregation: AVG

Unit: Percentage

Business Interpretation: Higher values indicate improved circular economy practices.

Typical Drill-down: Facility, Country

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Employee Count

Business Definition: Number of employees.

Recommended Aggregation: SUM

Unit: Employees

Business Interpretation: Represents workforce size.

Typical Drill-down: Country, Facility

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Female Workforce Percentage

Business Definition: Female representation in workforce.

Recommended Aggregation: AVG

Unit: Percentage

Business Interpretation: Measures workforce diversity.

Typical Drill-down: Country, Industry

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Training Hours

Business Definition: Training delivered to employees.

Recommended Aggregation: AVG

Unit: Hours

Business Interpretation: Reflects employee development.

Typical Drill-down: Country, Industry

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.

Compliance Violations

Business Definition: Recorded compliance incidents.

Recommended Aggregation: SUM

Unit: Incidents

Business Interpretation: Lower values indicate stronger governance.

Typical Drill-down: Country, Facility

Recommended Visualization: Bar

Example Business Questions: Compare across countries, identify top-performing facilities, analyze yearly trends, benchmark industries, and support executive ESG reporting.